

HematoVision: Advanced Blood Cell Classification Using Transfer Learning

HematoVision is a web-based deep learning project that uses transfer learning (VGG16) to classify human blood cells into five types: Neutrophil, Lymphocyte, Monocyte, Eosinophil, and Basophil. It is designed to assist medical professionals and labs by automating the process of blood cell classification from microscopic images.

Features

-  Predicts blood cell type from an uploaded image
-  Utilizes VGG16 pre-trained model (Transfer Learning)
-  Web interface built with Flask + HTML/CSS
-  Dockerized and ready for deployment
-  User authentication (login/register)
-  Model auto-downloaded via gdown from Google Drive

Project Structure

HematoVision/

```
├── app.py          # Flask application
├── Dockerfile      # Docker setup
├── requirements.txt # Required Python packages
├── static/
│   ├── css/
│   │   └── style.css
│   └── uploads/
├── templates/
│   ├── index.html
│   ├── result.html
│   ├── login.html
│   └── register.html
└── hemato_vision_vgg16.h5 # Trained model (downloaded via gdown)
```

⚙ Technologies Used

Frontend: HTML, CSS

Backend: Flask

Deep Learning: TensorFlow, Keras, VGG16

Image Processing: OpenCV

Deployment: Docker + Render

🩺 Blood Cell Types Detected

Neutrophil

Lymphocyte

Monocyte

Eosinophil

Basophil

📎 Download Pretrained Model

This project requires a trained CNN model file to make predictions.

👉 Download Blood_Cell_PRED.h5

Place the file in the root directory of the project (same location as app.py).

🚀 How to Run Locally

Step 1: Clone the repository

git clone <https://github.com/Renuka665/Hematovision>

cd Blood-Cell-Classification

Step 2: Run using Docker

docker build -t hematovision-app .

docker run -p 5000:5000 hematovision-app

Then open: <http://localhost:5000>

📦 Requirements

For manual installation (if not using Docker):

flask==2.3.3

tensorflow==2.10.0

opencv-python

gdown

werkzeug==2.3.7

numpy

pandas

Model File

The model Blood_Cell_PRED.h5 is automatically downloaded at runtime using gdown. You don't need to upload it manually.

Sample Use

Upload a microscopic image of a blood smear → The model predicts the cell type → You get a label and confidence score.

Authentication

Includes basic login and registration system using Flask sessions and SQLite for storing user details.

Results

- ✓ Training Accuracy: 96.6%
- ✓ Validation Accuracy: 67.8%
- ✓ Real-time prediction within 2 seconds

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License

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